CHAPTER 10 · REFERENCE DEPTH

Modulation

Your voice is a slow, weak vibration in the air. A radio wave is a fast, powerful oscillation that can leap across a continent. Modulation is the marriage of the two — the trick that lets a feeble 1 kHz voice ride on a 120 MHz carrier and arrive intact at a control tower. Get it right and you understand why aviation insists on AM for VHF and SSB for HF.

SYLLABUS MAP

Part II (iii) Systems for air-ground communication – carrier, AM/FM/SSB, the Tx/Rx chain

◎ Learning objectives — by the end of this chapter you will be able to...

- Explain what a carrier wave is and the two reasons modulation is needed.
- Describe AM in depth — sidebands, bandwidth, modulation depth and power distribution.
- Describe FM — deviation, bandwidth (Carson's rule), noise immunity and the capture effect.
- Compare DSB and SSB and calculate the power/bandwidth saving.

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- Explain why aviation uses AM on VHF and SSB on HF, and read emission designators.
- Outline the transmitter chain and the superheterodyne receiver.

10.1 The carrier & why we modulate

10.2 AM — sidebands & depth

10.3 AM power distribution

10.4 FM

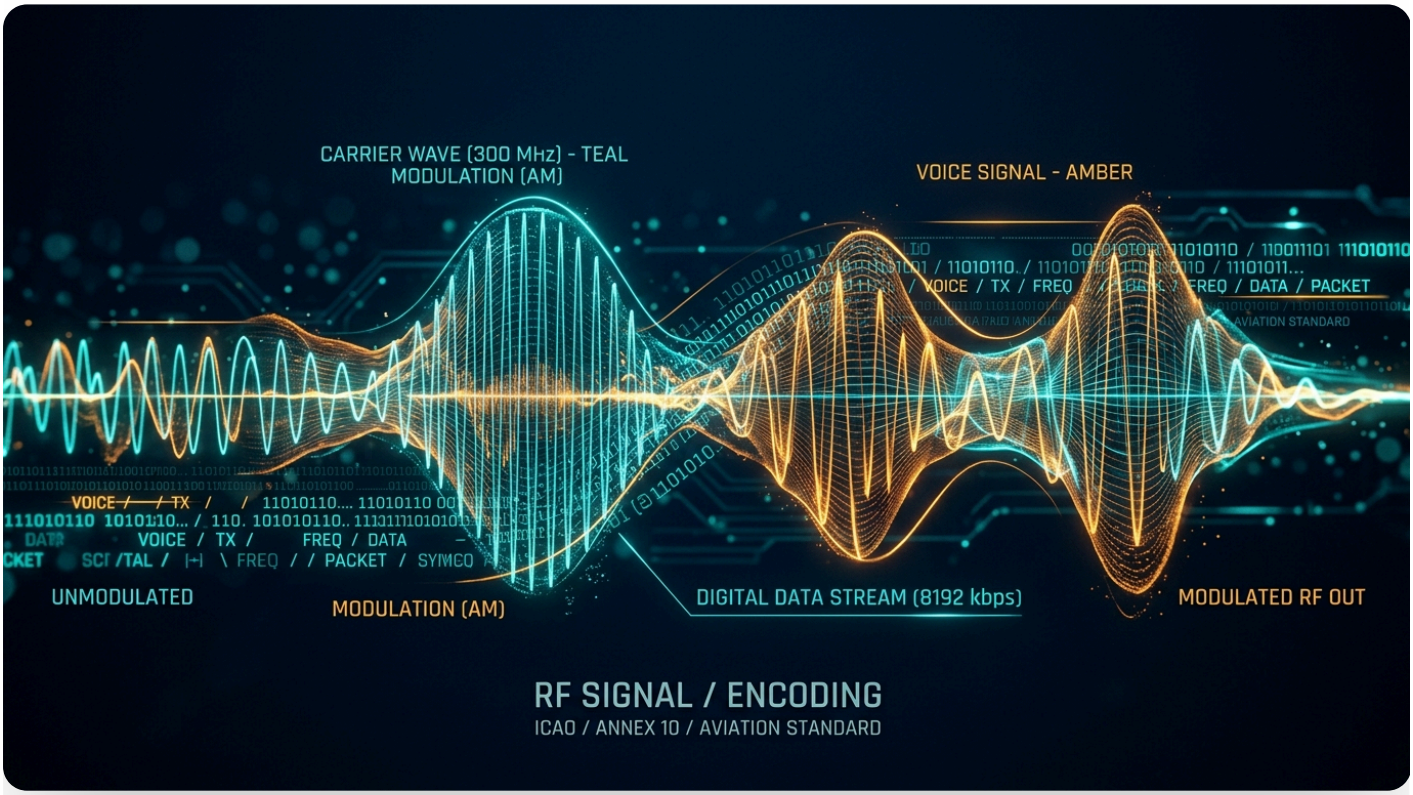
10.5 DSB & SSB

10.6 Why AM on VHF, SSB on HF

10.7 Emission designators

10.8 The Tx & Rx chains

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Modulation is the process of loading information—like your voice or digital data—onto a high-frequency carrier wave.

10.1 The carrier & why we modulate

FIRST PRINCIPLES — TWO REASONS WE CANNOT SEND THE VOICE DIRECTLY

You cannot simply connect a microphone to an antenna. **First**, an efficient antenna must be a fair fraction of a wavelength; a 1 kHz audio tone has a wavelength of 300 km, demanding an absurd aerial. **Second**, every voice would occupy the same low band, so no two stations could share the air. The solution is to put the voice onto a **high-frequency carrier**: the carrier needs only a short antenna, and each station can use a different carrier frequency. **Modulation** is the act of loading the voice onto the carrier.

DEFINITION — CARRIER & MODULATION

The **carrier wave** is a high-frequency wave generated by the transmitter; on its own it carries **no information**. **Modulation** varies one property of the carrier — its **amplitude** or its **frequency** — in step with the low-frequency audio, so the information rides on the carrier and can be recovered at the receiver.

10.2 AM — sidebands & modulation depth

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IN PLAIN TERMS

In Amplitude Modulation, the **height (amplitude)** of the carrier is varied in step with the voice, while its frequency stays fixed. The outline (envelope) of the wave traces your voice.

SIDEBANDS & BANDWIDTH

Mixing a carrier (fc) with an audio tone (fa) produces two new frequencies either side of the carrier: an **upper sideband** (fc + fa) and a **lower sideband** (fc − fa). The information lives in the sidebands; the carrier itself carries none.

The total AM bandwidth = 2 × highest audio frequency.

Modulation depth $m = V_m / V_c$

V_m = peak modulating (audio) voltage · V_c = carrier voltage · usually expressed as a %

MODULATION DEPTH (INDEX)

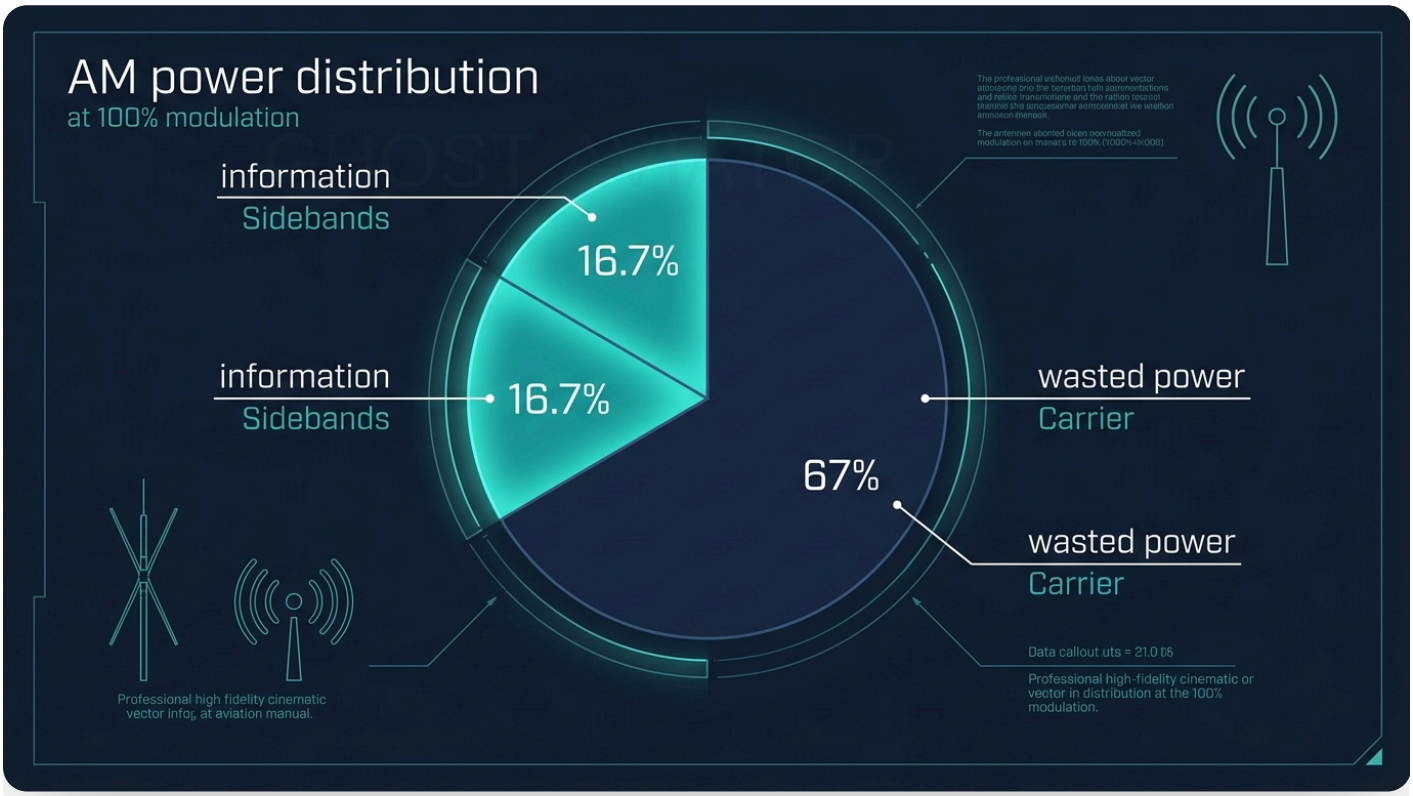
The **depth of modulation** is how strongly the carrier is varied. **Too little** (low %) wastes power and gives a weak, quiet signal; **over 100% (over-modulation)** distorts the audio and "splatters" energy into adjacent channels, causing interference.

10.3 AM power distribution

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WHY AM IS POWER-INEFFICIENT

Even at full 100% modulation, **two-thirds of the transmitted power sits in the carrier**, which carries no information; only one-third is shared between the two sidebands. That waste is the motivation for SSB.



At 100% modulation depth, ≈67% of the transmitter's power is wasted in the carrier, leaving only ≈16.7% for each information-carrying sideband.

WORKED EXAMPLE — WHERE THE POWER GOES

At 100% modulation, of the total power: **~67%** is in the carrier, **~16.7%** in each sideband. The information-bearing power is only the two sidebands **≈ 33%**.

SSB throws away the carrier and one sideband, putting essentially **all** the transmitter's

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power into the single information-bearing sideband — a large efficiency gain, decisive over long HF distances.

10.4 FM — Frequency Modulation

IN PLAIN TERMS

In FM the carrier's **frequency** is varied in step with the voice while its **amplitude stays constant**. The amount of frequency swing is the **deviation**. Because most noise affects amplitude, FM is far more noise-resistant — which is why it is used for music broadcasting.

FM BANDWIDTH — CARSON'S RULE

FM bandwidth ≈ **2 × (deviation + highest audio frequency)**. FM generally needs **more bandwidth** than AM, which is part of why narrow aviation channels stayed with AM.

⚠ EXAM TRAP — THE CAPTURE EFFECT

FM has a **capture effect**: the stronger of two signals captures the receiver and the weaker is silenced. In aviation that is dangerous — a second aircraft transmitting would be **silently lost. This is the key reason aviation does NOT use FM for VHF voice.**

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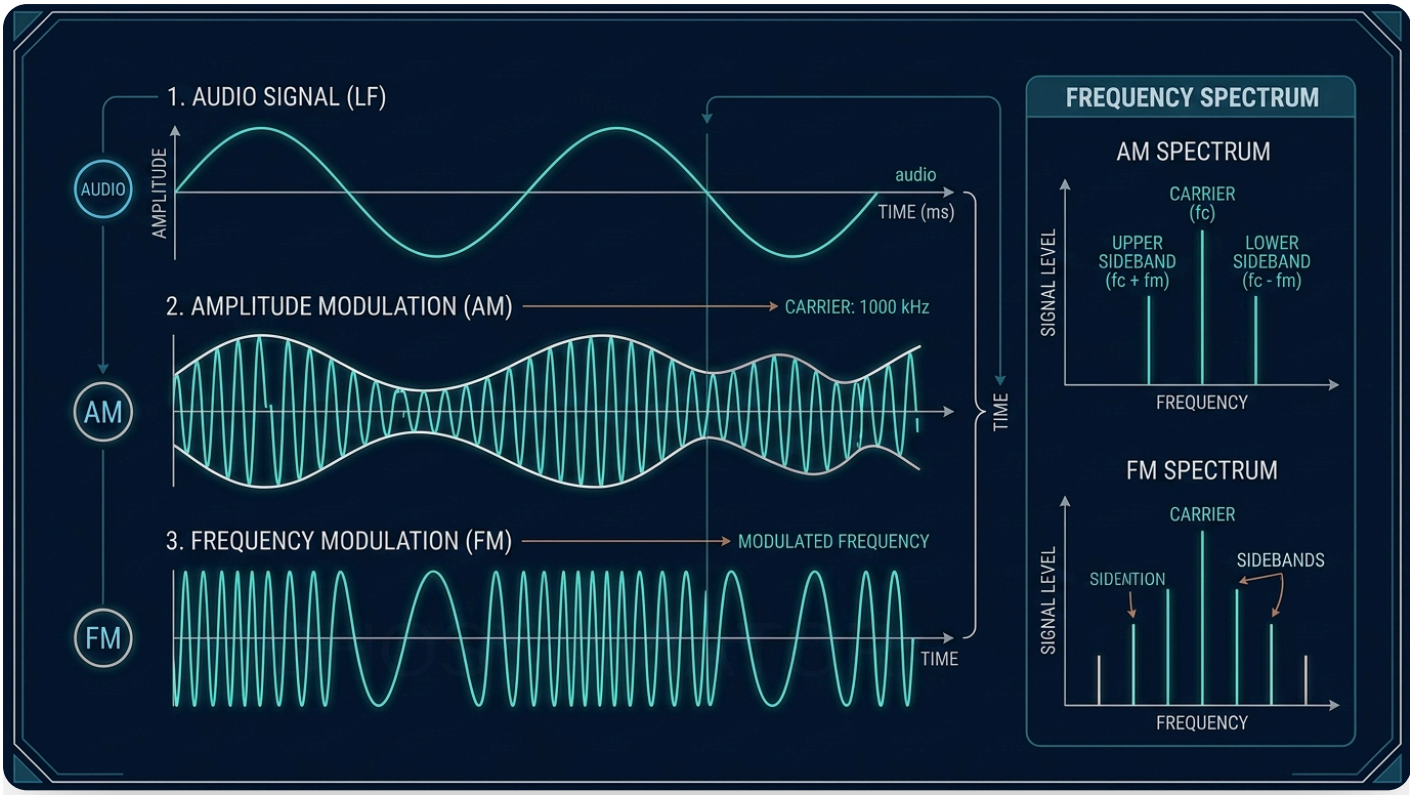


Figure 10.1 — AM vs FM waveforms, the sidebands, and the concept behind Single Side Band (SSB).

10.5 DSB & SSB

Modulation Modes Compared

Mode	What is transmitted	Trade-off
DSB (full AM)	Carrier + both sidebands	Simple receiver; but wasteful — most power in the carrier
DSB-SC	Both sidebands, carrier suppressed	Saves the carrier power; needs carrier reinsertion

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Mode	What is transmitted	Trade-off
SSB	One sideband only; carrier & other sideband suppressed	Far more power- and bandwidth-efficient — ideal for HF long range

10.6 Why AM on VHF, SSB on HF

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⚖ THE TWO DESIGN DECISIONS, SIDE BY SIDE

VHF voice = AM: if two aircraft transmit together on the same VHF frequency, AM produces an audible **heterodyne squeal** — everyone hears that two stations "stepped on" each other and can re-transmit. With FM's capture effect, one would simply be lost. AM's "you'll hear the clash" behaviour is a **safety feature**.

HF voice = SSB: over oceanic distances, efficiency and bandwidth matter most, so HF uses the power- and bandwidth-saving SSB.

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WHY SSB FOR HF

SSB puts all the transmitter's power into the one sideband that carries information and uses **half the bandwidth** of full AM. Over thousands of miles of HF that efficiency is decisive — so **HF aviation voice uses SSB**. The receiver must reinsert a local carrier to demodulate it.

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✈ COCKPIT REALITY

That brief squeal you sometimes hear after a call is two aircraft transmitting together. Because VHF is AM, you know it happened and can say "say again, two together." That awareness is exactly what AM buys aviation — and what FM would hide.

10.7 Emission designators

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</> THE ICAO/ITU THREE-SYMBOL CODE

An emission is classified by three symbols: **type of modulation · nature of the signal · type of information**. The two you must recognise:

A3E = amplitude-modulated double-sideband telephony (aviation VHF voice)

J3E = single-sideband suppressed-carrier telephony (aviation HF voice).

10.8 The transmitter & receiver chains

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TRANSMITTER — VOICE TO AIRWAVES

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Microphone → audio amplifier → **oscillator** (generates the carrier) → **modulator** (mixes audio onto the carrier) → **RF power amplifier** → antenna.

RECEIVER — AIRWAVES TO VOICE (SUPERHETERODYNE)

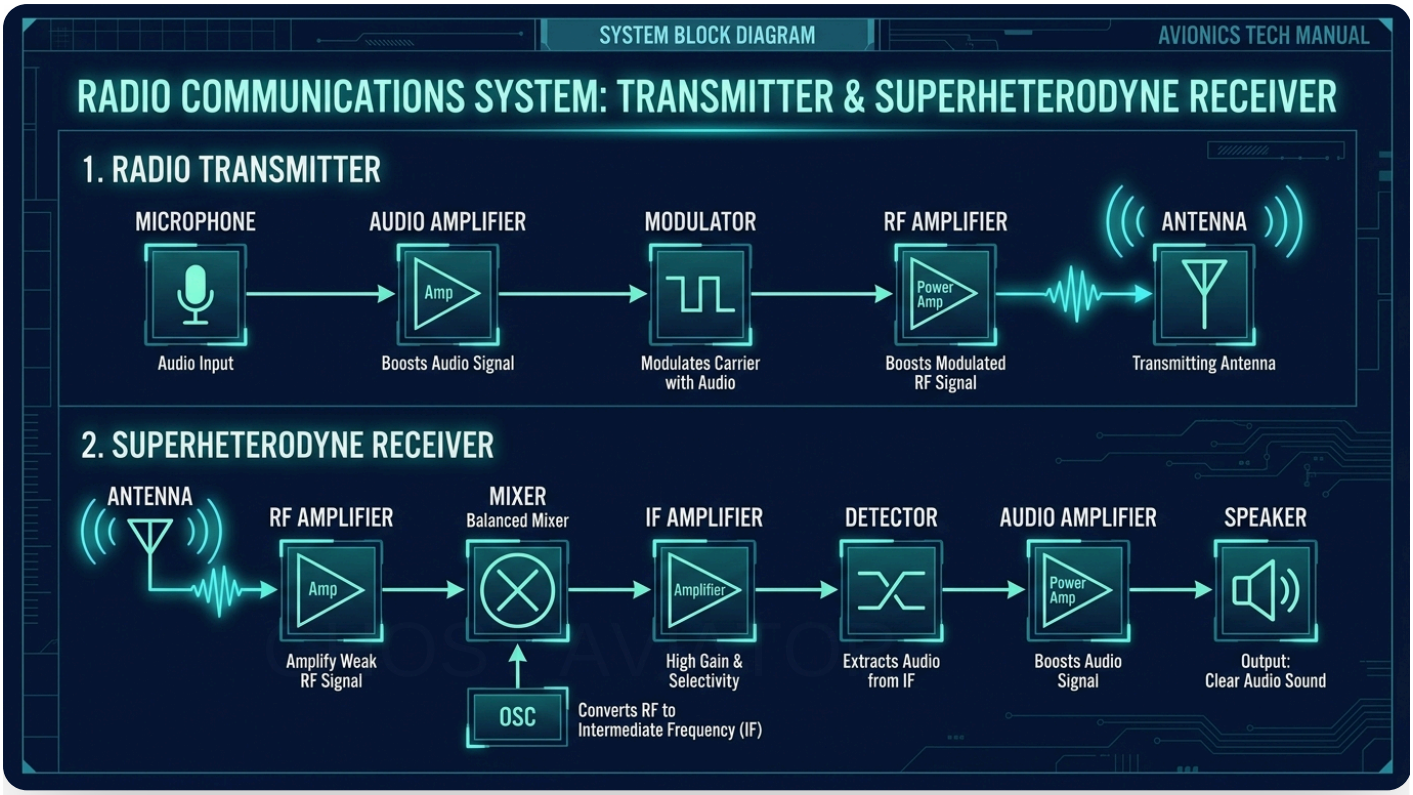
Antenna → RF amplifier → **mixer + local oscillator** (shifts the signal to a fixed **intermediate frequency**, IF) → IF amplifier (most gain/selectivity) → **detector / demodulator** (recovers the audio) → audio amplifier → speaker/headset. **AGC/AVC** holds the output level steady (Chapter 11).

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WHY "SUPERHETERODYNE"

Converting every incoming signal to one fixed IF means the gain and selectivity stages are always tuned to the same frequency — far easier to make highly selective than tuning every stage to each station. It is the architecture of virtually every aviation receiver.

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The essential block diagrams of a radio transmitter and a superheterodyne receiver.

☆ Numbers to memorise

Essential Facts for Chapter 10	
Fact	**Value**
Carrier	High-frequency wave; carries no info until modulated
AM	Amplitude varied; bandwidth = 2 × highest audio; info in sidebands
Modulation depth	m = Vm/Vc; over 100% → distortion & splatter

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Fact	Value
AM power (100%)	≈ 67% carrier, ≈ 33% sidebands (the only useful part)
FM	Frequency varied; noise-resistant; capture effect; Carson BW = 2(dev + fmax)
SSB	One sideband, carrier suppressed; efficient; HF voice
Aviation choice	VHF voice = AM (A3E) · HF voice = SSB (J3E)
Receiver	Superheterodyne — converts to a fixed IF

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🖉 Question bank

Part A — MCQs (click an option to check)

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1. In amplitude modulation, what is varied?

The amplitude of the carrier

The frequency of the carrier

The speed of light

The antenna length

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Answer: The amplitude of the carrier.

AM varies the carrier's amplitude in step with the audio; frequency stays fixed.

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2. One reason a voice cannot be transmitted directly is that it would require:

Too little power

An impractically long antenna (its wavelength is enormous)

A satellite

Frequency modulation

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Answer: An impractically long antenna (its wavelength is enormous).

Audio wavelengths are hundreds of km; modulating a high-frequency carrier allows a short antenna and frequency sharing.

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3. The information in an AM signal is carried in the:

Carrier

Sidebands

Antenna

Local oscillator

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Answer: Sidebands.

The carrier carries no information; the sidebands do.

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4. If the highest audio frequency is 3 kHz, the AM bandwidth is:

3 kHz

6 kHz

1.5 kHz

9 kHz

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Answer: 6 kHz.

AM bandwidth = 2 × highest audio frequency = 2 × 3 = 6 kHz.

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5. Modulating beyond 100% depth causes:

A clearer signal

More range

Distortion and interference to adjacent channels

Lower frequency

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Answer: Distortion and interference to adjacent channels.

Over-modulation distorts the audio and splatters into neighbouring channels.

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6. At 100% AM modulation, the proportion of power in the carrier is about:

33%

67%

100%

0%

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Answer: 67%.

About two-thirds of the power is in the carrier, which carries no information.

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7. In FM, what is held constant?

Frequency

Amplitude

Wavelength of the audio

Bandwidth

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Answer: Amplitude.

FM varies frequency while amplitude stays constant — hence its noise immunity.

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8. FM is generally NOT used for VHF aviation voice because of its:

Low cost

Capture effect (weaker signal silenced)

Wide range

Use of sidebands

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Answer: Capture effect (weaker signal silenced).

FM's capture effect would silently suppress a second transmitting aircraft — unsafe.

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9. Aviation VHF voice uses AM mainly because:

It is cheaper

Two simultaneous transmissions produce an audible heterodyne, alerting users

It has the longest range

It needs no antenna

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Answer: Two simultaneous transmissions produce an audible heterodyne, alerting users.

AM reveals when two stations transmit together (a squeal); FM's capture effect would silently lose one.

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10. SSB is preferred for HF because it is:

Simpler to receive

More power- and bandwidth-efficient

Immune to all fading

Higher frequency

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Answer: More power- and bandwidth-efficient.

SSB suppresses the carrier and one sideband, concentrating power and halving bandwidth.

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11. SSB uses how much bandwidth compared with full AM?

Twice

Half

The same

Four times

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Answer: Half.

Transmitting one sideband instead of two halves the bandwidth.

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12. The emission designator for aviation VHF voice (AM telephony) is:

J3E

A3E

F3E

A1A

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Answer: A3E.

A3E = AM double-sideband telephony (VHF voice); J3E = SSB telephony (HF voice).

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13. The aviation receiver architecture that converts every signal to a fixed intermediate frequency is the:

Direct-conversion receiver

Superheterodyne receiver

Crystal set

Regenerative receiver

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Answer: Superheterodyne receiver.

The superheterodyne mixes the signal down to a fixed IF for easy selectivity and gain.

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14. In the transmitter chain, the stage that generates the carrier is the:

Modulator

Oscillator

Power amplifier

Antenna

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Answer: Oscillator.

The oscillator generates the carrier; the modulator then mixes the audio onto it.

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15. Modulation depth m is defined as:

Carrier voltage ÷ audio voltage

Audio (modulating) voltage ÷ carrier voltage

Frequency ÷ wavelength

Power ÷ bandwidth

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Answer: Audio (modulating) voltage ÷ carrier voltage.

m = Vm/Vc, usually expressed as a percentage.

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16. Compared with AM, FM generally requires:

Less bandwidth

More bandwidth

No bandwidth

The same bandwidth

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Answer: More bandwidth.

By Carson's rule, FM bandwidth ≈ 2(deviation + fmax), usually more than AM.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **What is modulation, and why is it needed?**

Model Answer:

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Modulation varies a property of a high-frequency carrier — its amplitude or frequency — in step with the low-frequency audio. It is needed because audio wavelengths are too long for a practical antenna and every voice would occupy the same band; a high-frequency carrier allows a short antenna and lets stations share the spectrum.

▼ ? **Describe AM, its sidebands and bandwidth.**

Model Answer:

In AM the carrier's amplitude follows the audio. Mixing produces upper and lower sidebands

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either side of the carrier; the information is in the sidebands. The bandwidth is twice the highest audio frequency.

▼ ? Why does aviation use AM for VHF voice?

Model Answer:

With AM, two aircraft transmitting simultaneously produce an audible heterodyne squeal, so everyone knows a clash occurred and can re-transmit. FM's capture effect would silently suppress the weaker signal — a safety hazard.

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▼ ? Why is SSB used for HF?

Model Answer:

SSB suppresses the carrier and one sideband, putting all the power into the single information-bearing sideband and using half the bandwidth — far more efficient over long HF distances.

▼ ? What is over-modulation and its effect?

Model Answer:

Modulating beyond 100% depth, which distorts the transmitted audio and causes splatter — interference into adjacent channels.

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▼ ? Outline the superheterodyne receiver.

Model Answer:

The signal is amplified, then mixed with a local oscillator to a fixed intermediate frequency, amplified at the IF (where most gain and selectivity occur), demodulated to recover the audio, then amplified to the speaker, with AGC holding the level steady.

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Part C — Numerical problems (tap for worked solutions)

▼ 🧮 **P1. A carrier of 120.000 MHz is amplitude-modulated by a 2.5 kHz tone. Give the sideband frequencies and bandwidth.**

Solution:

USB = 120.0000 + 0.0025 = **120.0025 MHz**; LSB = **119.9975 MHz**;

bandwidth = 2 × 2.5 = **5 kHz**.

▼ 🧮 **P2. A carrier of 10 V is modulated by a 6 V audio signal. Find the modulation depth.**

Solution:

m = Vm/Vc = 6/10 = 0.6 = **60%** (within limits).

▼ 🧮 **P3. An FM signal has a deviation of 5 kHz and a top audio of 3 kHz. Estimate its bandwidth.**

Solution:

Carson's rule: BW ≈ 2(5 + 3) = **16 kHz**.

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⚡ 60-SECOND REVISION CARD

Carrier = blank vehicle; modulation loads the voice (needed for a short antenna + frequency sharing).

AM varies amplitude; **BW = 2 × fmax**; info in sidebands; ≈67% power wasted in the carrier.

FM varies frequency, noise-resistant, wider bandwidth, has the **capture effect**.

SSB = one sideband, carrier suppressed, half the bandwidth → **HF (J3E)**; **AM = VHF voice (A3E)**.

Over-modulation (>100%) → distortion & splatter. Receiver = **superheterodyne** (fixed IF).

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